SK-2N01G Shale Density Measurement

Standard Operating Procedure | Sample preparation, zeroing, measurement, calculation and quality control

1 Scope

This SOP describes manual shale or mudstone cuttings density measurement using the SK-2N01G hydrometer-type shale density meter. It is intended for trained mud logging or wellsite geology personnel.

2 Principle

The same prepared specimen is measured first on the upper pan in air and then on the submerged support. The change in response is caused by water buoyancy and is used to calculate density.

$$\text{Density (g/cm}^3\text{)} = L1 / (L1 - L2)$$

Item	Specification
L1	Stable scale reading with the specimen on the upper pan in air
L2	Stable scale reading with the same specimen on the submerged support
Validity condition	L1 must be greater than L2 and (L1 - L2) must be positive

3 Responsibilities

- Operator: prepare representative samples, inspect and zero the instrument, perform readings, calculate and record results.
- Mud logging geologist / supervisor: define sampling interval, review quality-control results and approve trend interpretation.
- Wellsite supervisor: ensure results are integrated with the approved formation-pressure and well-control workflow.

4 Required equipment and materials

- Complete SK-2N01G instrument, transparent cylinder and protective cover.
- Clean fresh water at stable room temperature.
- Sample sieve, tweezers, soft brush, wash bottle and absorbent lint-free paper.
- Level, stable work surface; sample tray; depth labels; worksheet or approved digital record.
- Gloves and eye protection required by the site risk assessment.

5 Safety and quality precautions

- Handle cuttings as potentially contaminated with drilling fluid and hydrocarbons. Follow the site's chemical hygiene and waste-disposal requirements.
- Use clean fresh water only for this two-reading procedure unless the approved site method specifies a different fluid and correction.
- Do not force, bend or drop the rod, float, scale or stabilizing weight. Do not operate with damaged or sticking parts.
- Do not interpret one measurement as proof of abnormal pressure. Escalate anomalous trends through the approved well-control process.

6 Procedure

Step 1 - Verify the sample interval

Confirm sample depth, lag correction and lithology. Reject samples with uncertain depth or obvious contamination. Use competent shale or mudstone representative of the drilled interval.

Step 2 - Select the specimen

Select intact, angular pieces of comparable size. Avoid cavings, highly porous or gumbo material, flat flakes, pieces with visible cracks or voids, and fragments containing obvious sand, carbonate, pyrite or other non-shale material unless these are deliberately being characterized.

Step 3 - Wash and blot

Wash the specimen gently to remove adhering drilling fluid without abrading or hydrating it unnecessarily. Blot surface liquid with absorbent paper. Do not oven-dry, heat or leave the sample exposed long enough to change its moisture condition. Measure promptly.

Step 4 - Set up the instrument

Place the cylinder vertically on a stable, vibration-free bench. Fill with clean fresh water. Install the rod, upper pan, float / submerged support and stabilizing weight. Adjust the water level so the float operates within the scale and the moving assembly has clearance from the cylinder wall and bottom. For the conventional setup, the float top is approximately 1 cm below the water surface; use the delivered instrument's reference mark where provided.

Step 5 - Inspect and zero

Remove all specimens. Confirm free movement, no trapped bubbles, and no contact with the cylinder. Allow the assembly to stabilize, then use the zero-adjustment control to set the empty instrument to zero. Recheck zero whenever room or water temperature changes materially.

Step 6 - Obtain L1

Place the prepared specimen on the upper pan. Wait until oscillation has stopped and the stabilizing weight remains clear of the bottom. Read the scale at eye level and record the stable value as L1.

Step 7 - Obtain L2

Using tweezers, move the same specimen to the submerged specimen support. Dislodge visible air bubbles without losing material. Wait for stabilization, verify that no component touches the cylinder, and record the stable value as L2.

Step 8 - Calculate density

Calculate Density = L1 / (L1 - L2). Record L1, L2, the calculated density, sample depth, lithology, sample condition, date, time and operator. Do not report a value if L1 ≤ L2 or if the denominator is zero or unstable.

Worked example

If L1 = 2.40 and L2 = 1.40, then Density = 2.40 / (2.40 - 1.40) = 2.40 g/cm³. This example demonstrates the calculation only; actual scale readings depend on the instrument.

7 Replicates and quality control

- Measure at least three representative pieces per interval; five pieces are preferred when sample quantity permits.
- Use the same washing, blotting, settling and reading timing for all specimens.
- For five specimens, review the full spread; a common robust practice is to discard the highest and lowest values and average the middle three, provided there is no evidence of mixed lithology or procedural error.
- Repeat a measurement when bubbles, wall contact, bottom contact, unstable zero, sticking movement or abnormal sample behavior is observed.
- Use the site's approved repeatability limit. If no limit exists, retain all readings and flag excessive scatter for supervisor review rather than silently selecting a preferred result.

8 Result review

- Confirm calculated values lie within the instrument's nominal 1.00-3.00 g/cm³ range.
- Compare the result with adjacent intervals and lithology descriptions. Investigate abrupt changes before accepting them.
- Plot accepted interval values against corrected sample depth and maintain the established normal compaction trend.
- Treat a downward departure or reduced rate of density increase as an indicator requiring integrated review, not as a stand-alone pressure determination.

9 Cleaning and storage

Remove all sample fragments. Drain the cylinder and rinse the cylinder, pans, float, rod and stabilizing weight with clean water. Use a soft cloth or brush; do not use abrasives. Dry components before assembly or storage. Store the instrument upright or in its protective case with the rod and scale protected from bending and impact.

10 Troubleshooting

Item	Specification
Cannot set zero	Check assembly order, water level, trapped bubbles, deposits, bent parts, wall/bottom contact and bench vibration.
Reading will not stabilize	Remove bubbles; confirm free movement; isolate vibration and drafts; verify the sample is secure and the stabilizing weight is clear.
L2 ≥ L1	Reading order, specimen position or scale reading is incorrect. Reset, re-zero and repeat.
Results show large scatter	Review sample representativeness, mixed lithology, cracks, hydration, contamination, blotting consistency, bubbles and operator timing.
Trend changes abruptly	Repeat the interval, inspect sample depth and lithology, then compare with independent drilling and formation-pressure indicators.

11 Minimum record

Item	Specification
Well / hole	_____
Sample depth / interval	_____
Lag-corrected depth	_____
Lithology and sample condition	_____
L1	_____
L2	_____
Calculated density	_____ g/cm³
Replicate results / average	_____
Operator / date / time	_____
QC observations	_____

12 Acceptance checklist

- Correct model and complete undamaged assembly verified.
- Representative depth and shale / mudstone lithology confirmed.
- Instrument clean, free-moving and zeroed before measurement.
- No trapped bubbles or wall / bottom contact during readings.
- L1 and L2 recorded; calculation independently checked.
- Replicates and QC comments retained; result reviewed in context.

References and control

Procedure basis: published SK-2N01G operating guidance and established mud logging shale-density quality practices. Site-specific instructions, delivered-equipment markings and the operator's well-control program take precedence.

References: https://www.renrendoc.com/paper/105970154.html | https://imudlogging.com/product/iml-3dt01-shale-density-meter | https://www.chinamudlogging.com/product/shale-density-meter-for-mud-logging-unit/

SOP revision: 1.0 | Effective date: 2026-10-14 | Review before field issue